CS-013 Work Orders & Part Traceability

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Title	Work Orders & Part Traceability
Revision	G
Status	Draft, pending Lead signature
Effective date	2026-10-10
Supersedes	Master Tracker part rows as the only part record

Approvals

Role	Name	Date
Author	Simon Starbuck (drafted with Claude, SN6 Resources)	2026-07-12
Reviewer	simon (reviewer agent)	2026-10-10
Approver (Composites Lead)		

Revision history

Rev	Date	Author	Description of change
A	2026-07-12	Claude	Initial issue, defines the WO record and its authority rules
B	2026-07-28	Claude	Corrected the app description throughout: it's the live Firebase app now, not `work-orders.html`; buy-off, Backup/Restore, and photo-attachment wording updated to match. Language pass (fewer em dashes, less decorative bold).

Rev	Date	Author	Description of change
C	2026-08-03	Claude	Added §4.1, the ID grammar, which had no controlling document: the prefix table, the rule that an ID names a physical object and never a job, non-reuse, and `supersedes` for a re-laid part. Molds, material lots, test panels and storage locations became real records, so §7.2 now points `moldRef` at a mold record rather than only at a free-text string. §7.2 also gains lot capture at the cure buy-off.
D	2026-08-26	Claude	Defined the R&D part / R&D WO in §4 and added the marking step to §7.1. R&D is a property of the object and orthogonal to status, so it does not join the status enum and does not displace the retro sentinel — the two can be true of the same record, and unlike a retro record an R&D one still enforces. §8 gains the check that catches an unmarked R&D build. The ID grammar is unchanged: R&D lives in a field, never in an ID, because the ID must survive a build being moved into the season.
E	2026-08-28	Claude	Engineering pass (CS-000 Rev C conventions). Figure 1 draws the record graph (which record points at which, and which rules keep the pointers true) because §4.1's four rules are relationships, and relationships read better drawn. No rule changed.

Rev	Date	Author	Description of change
F	2026-08-29	Claude	Coupons caught up with the app's v4.0.0 R&D bench. A coupon is its own record, `CPN-SNx-###`, under a study `RDS-SNx-###`; the suffix spelling `PNL-SNx-###-Cnn` is retired and survives only on frozen SN5-era records. §2 and §4 now separate bench coupons (no work order, no cure hold) from R&D parts (full enforcement), which the standard previously conflated. Figure 1 redrawn to match.

Rev	Date	Author	Description of change
G	2026-10-10	Claude (doc overhaul, requested by Simon Starbuck)	Caught up with the app since v4.1. A re-laid part is a new run of the same part, not a new P ID (§4.1 rule 3); runs start from the part (§7.1); STK, BUY and DOC join the prefix table. Rules written for what the app already gates: training on buy-offs, independent mold design review, cure holds and evidence, issue dispositions before Complete. New §7.5 techniques (with the built-in checklists and a shall to keep the Stack frozen and drop-test blockers), §7.6 issues, §7.7 archive and Recently deleted. Accounts, photos, lots, molds, R&D, person fields and the weekly board match the app; study versus part in §2; two §8 rows. Register R-013-01..23, R-013-30 (blocker and hold, the app's words, in place of "gate"), R-002-08, R-002-13, R-000-09, R-000-11. The self-review warning is described as the app now keys it, on the run's Mold Engineer (R-OTH-32).

1. Purpose

The Master Tracker did its job at the fleet level: parts, owners, deadlines, all visible. What it never gave us was the *part-level* record: steps, buy-offs, the stack as actually laid, measurements, photos. All of that lived in Slack threads and people's heads (PP-09). A work order (WO) is one run's complete record (a part has one run per attempt at making it, §4.1 rule 3), spec through blockers through execution through evidence, on a screen or printed and kept with the part.

2. Scope

Every manufactured composites part and mold, including small cross-team jobs (the catch-can class of work, PP-05). If it consumes carbon or machine time, it gets a WO, with one carve-out: coupon-scale test pieces live on the app's R&D bench as studies and coupons (`RDS` / `CPN`, §4.1) with no work order, no traveler and no cure hold. The panel a coupon is cut from is still a build under this standard; the pieces cut from it are bench records.

Which record to make: test pieces you will cut, measure and compare (coupons, offcuts, trial panels you will not keep) should go in a study. Anything with a mold, a checklist or a cure you need to sign should be a part and a run, marked R&D if it is not for the car (§4). A study may also hold the parts and runs an effort produced, as a folder. Filing a part under a study does not make it R&D, and deleting a study ungroups its parts without deleting them.

3. References

Ref	Document	Where
R1	PP-09, PP-05, PP-03 RCAs	FEB-COMP-PP-001, `01 Pain Points and Improvements/src/pain-points.md`
R2	The work-order app (live, Firebase-backed)	`06 Composites App/app/README.md`, live at feb-composites.web.app
R2b	`work-orders.html`, the original offline single-file version, kept as a backup/archive viewer only, not the primary system	`06 Composites App/work-orders.html`
R3	CS-002 (stack), CS-003 §7.2 (design review), CS-006/007 (process records), CS-005 §4 (ShopSabre operator authority), CS-008 §5.1 (team cure holds)	this series
R4	CS-000 (revision rules apply to WOs)	CS-000

4. Definitions

- **WO ID**: `WO-SNx-###`, assigned from the app (or the next free number in the printed log). Never reused.
- **Part and run**: a part (`P-SNx-###`) is the durable thing the car needs. A run is one attempt at making it; the app's Work Orders tab calls a run a work order. A part has many runs, and each run belongs to one part (§7.1).
- **ME / RE**: the part's Mold Engineer (ME) and Manufacturing Engineer (RE), the two engineer fields on the part and on each of its runs. Both are roster picks (Person fields, below). Same definition as CS-002 §4.
- **Person fields**: every field that names a person (Mold Engineer, Manufacturing Engineer, Sealed by, Laid up by and the rest) is picked from the app's roster, not typed. Pick Use "name" only for someone who is not on the app.
- **Buy-off**: pressing a step's buy-off while signed in. The app stores the signer's name as it read at signing, their account, and the date and time; screens show the person's current roster name. Self-buy-off is allowed except on Mold design review (below). It isn't tamper-proof: any roster member can still edit any record, so treat it as "who was signed in when this was clicked," not a notarized signature.
- **Training to sign**: a step that names a training (the §7.5 table shows which built-in step needs which) can be bought off only by someone a lead has recorded as holding it on the People tab. A lead may

sign without it, and must write why; the reason goes in the run's event log. A lead is not trained by being a lead. For the ShopSabre, the app's ShopSabre CNC training mirrors the RFS record, which CS-005 §4 names as the authority.

- **Independent design review**: Mold design review shall be bought off by someone other than the mold's designer, the part's Mold Engineer (CS-003 §7.2). The app only warns: when the run's Mold Engineer signs it, it asks "Self-review" ("You are this run's Mold Engineer…") with Sign it anyway, and it does not refuse. It reads only the Mold Engineer field, so the signer still checks who actually drew the CAD.
- **Blocker step**: a step that blocks everything after it until it's bought off (stack freeze, design review, drop test, and the acceptance criterion that opens the Other technique).
- **Cure hold**: the wait on the step after the one that starts a cure (Cure and demould, or Cure), as long as the team hold for the resin recorded at that buy-off (CS-008 §5.1). A member is refused with "Still curing". A lead can sign through it from "Demould early?" with a written reason, which goes in the event log.
- **OnHold** (run status): a run a person has parked. It blocks nothing in the app.
- **Evidence**: what a step needs before it can be signed. Stack frozen needs a layup stack; Mold design review needs the CAD, attached or linked; Machine mold, Glue and machine mold, the drop test and the acceptance criterion need a note on the step. A lead can Sign without it, with a written reason.
- **Retro WO**: a back-filled historical record (all `WO-SN5-*`); its unverifiable fields say "not recorded (retro)"; retro records are honest references, not exemplars of process compliance.
- **R&D part / R&D WO**: a build that feeds a decision rather than the car: coupon campaigns, flat test panels, layup trials, sealing and grounding experiments, mold shakedowns. An R&D build is real carbon, real machine time, real cost and a real deadline, with a full record under this standard.

An R&D part is not a bench coupon: a `CPN` record has no work order and enforces nothing (§2), while an R&D part keeps every blocker, hold and evidence check. R&D is a property of the object, not a status: a work order is R&D on the day it is a Draft and still R&D when it is Complete, and an R&D work order enforces its blockers, cure holds and evidence exactly like any other.

What an R&D part is not is a season deliverable. It is kept off the Season tab, the Google Sheet mirror, and the Parts and Work Orders lists; it lives on the R&D tab, and each of those two lists shows an R&D count that jumps there. It is counted and marked everywhere else, including the Dashboard and Reports, and marked `R&D` on screen, on the label (CS-001 §7.6), on the printed traveler and on the public scan page. The flag lives on the part; a work order inherits it from the part it builds, and a work order with no part carries its own.

4.1 The ID grammar

Until Rev C the only controlled identifier in the whole system was the WO ID. A mold was free text inside one work order, a material lot was the colour of its packaging, and a tensile coupon was a trailing integer in a filename. This is the grammar for all of it.

```
PREFIX-SNx-###          e.g. MOLD-SN6-004, P-SN6-007, CPN-SN6-042
```

Prefix	What it names
`WO`	a work order (the job)
`P`	a part
`MOLD`	a mold
`PNL`	a flat test panel
`BRD`	a tooling-board sheet or offcut
`FAB`	a fabric roll or usable offcut
`RSN`	a resin or hardener lot
`CON`	a consumable lot
`JIG`	a jig or fixture
`BIN`	a storage location

Prefix	What it names
RDS	an R&D study: a named set of coupons. Labels the bag or tray they live in
CPN	one coupon: a single physical test piece, belonging to a study
PROJ	an issue (older records say ticket). Not a physical object, and never gets a printed label (CS-001 §4).
STK	a mold's stack plan: how the mold is cut from tooling board, shown as the Mold file. Not a physical object
BUY	a purchase. Not a physical object
DOC	a document on the Documents tab. Not a physical object

Four rules:

1. **An ID names a physical object, never a job.** The one exception is WO, which is the job. Encode a work order into a part's ID and every re-lay either lies or forces a relabel. A P record names the part the car needs; each physical piece of it is told apart by the run that made it (rule 3). PROJ, STK, BUY and DOC name records that are not objects at all, and none of them gets a label.
2. **IDs are never reused**, across seasons or after a delete. The season segment (SN5, SN6) partitions the namespaces, so SN5 IDs are frozen and stay valid forever.
3. **A re-laid part is a new run, not a new part.** Scrap a layup for porosity and lay it again: dispose the issue on the failed run as Scrap, saying why it could not be saved (§7.6), then start a second run on the same part with + New run. The part keeps its P ID, and the run ID tells the two pieces of carbon apart. The mold's ID, the board's ID and every existing label are untouched.
4. **A remachined mold keeps its ID.** It is the same physical object at a new revision, so the revision letter changes and the label is reprinted; the ID does not move.

An ID is capped at **14 characters** by the QR code: version 3 at 25% error correction holds 47 alphanumeric characters, of which the host takes 30 and /Q/ takes 3. Everything above fits; MOLD-SNx-### at 12 characters is the longest. The retired coupon spelling PNL-SNx-###-Cnn was 15, one over budget, which is why those labels were text-only; it survives only on frozen SN5-era records (rule 2), and tools/test_qr.mjs keeps the 15-character form as a counterfactual so the cliff stays proven.

Dates are **fields, never ID segments.** A session that starts 23:50 and ends 00:10 has two valid dates, and a date typed wrong into an ID can only be corrected with acetone, while a date typed wrong into a field is corrected with an edit.

5. Equipment & Materials

The WO app (R2): a shared workspace at feb-composites.web.app, running on Firebase. Make an account on the sign-in page with New here? Create account: your name (it goes on your buy-offs), a username and a password. You start as a member and can work straight away. A lead makes someone a lead on the People tab, and removes someone who has left with Remove there; their name stays on every record they signed. View as guest reads the whole app and changes nothing, because every buy-off carries a name. Every edit saves live to a shared database, so the whole team sees the same record at once; there's nothing to import or merge. work-orders.html (R2b) still opens any exported JSON with no server at all, and stays useful as an offline archive viewer, but it's not where day-to-day work happens anymore. Printed WOs: any printer + a pen at RFS.

6. Safety

No physical hazards; process standard. (Safety content lives in the process standards each WO step cites.)

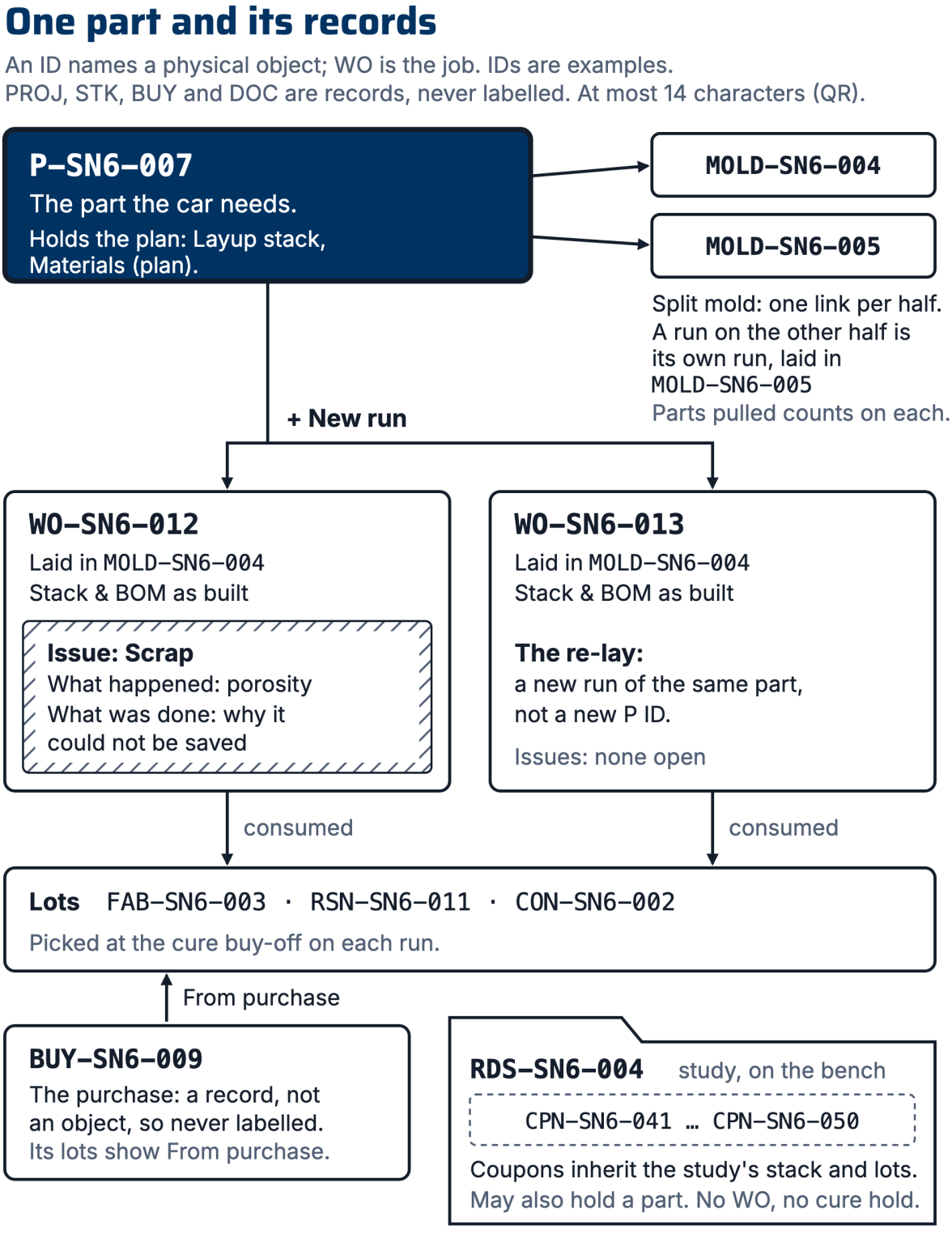


Figure 1: The record graph for one job: the work order builds a part, points at a mold record, and consumes lots; a study's coupons are records of their own, inheriting the study's stack and lots. The rules under the boxes are §4.1's, drawn where they apply.

7. Procedure

7.1 Create

1. Make the part on the Parts tab with New Part (it gets `P-SN6-###`), or open the part if it exists.
2. On the part, fill the requirements before any work (this is the PP-04/PP-05 fix: requirements exist before work): Subteam, Layup type (which picks the technique, §7.5), Layup deadline, Mold Engineer and Manufacturing Engineer, Target weight (g), the Layup stack (per CS-002 §7.1) and Materials (plan).
3. Press + New run on the part. The run gets `WO-SN6-###` and copies the part's plan: engineers, deadline, mass target, technique, stack, BOM and the part's first mold. For a remake, choose Use a previous run as the starting point and tick what to carry; Start fresh copies the part's own plan.
4. On the run, set the quality checks with numeric targets in Quality (+ check). Mass is prefilled from the part's Target weight.
5. The run's steps come from its technique. If the checklist does not fit the job, fix the technique, not the run: pick the right Layup type before the run starts, or ask a lead to add a technique (§7.5). Members cannot delete steps. A step that does not apply is signed with a note saying why.
6. Cross-team parts: the requesting subteam's requirements (thickness, DXF rev, mass) go in the WO and their lead is named on the release sign-off.
7. If the build feeds a decision rather than the car, start it from the R&D tab with + Part or + Run (§4). A run started on an R&D part is R&D with nothing to remember. Until a part has a run, the R&D part checkbox on its page (in edit mode) can be flipped either way. After that, only a lead can press Move to season, once, and nothing moves a season part back. The part keeps its ID when it moves (§4.1 rule 2), so its label, its traveler and its QR all stay valid.

New WO on the Work Orders tab makes a run with no part. Use it only for a one-off job that will never be remade.

7.2 Execute

1. Work the steps in order; blocker steps block until bought off (the app enforces this; on paper the blocker rows are shaded: no initials, no moving on).
2. Each step: buy-off (§4), a note, and photos. Press the camera on the step row (Add photos to this step) to attach photos straight from the phone; they land on that step and in the run's Photos section. Steps that need a note (machining, the drop test) also suggest a photo before signing; a photo is never required.
3. Deviations from the frozen stack or spec are a run revision, done as CS-002 §7.2 steps 4 to 6 say: quick, visible, counted.
4. **Which lots went in.** Buying off an infusion or a wet layup asks which fabric roll, which resin lot and which hardener lot, alongside the resin system and the finish time it already asked for. Each field is pre-filled with whatever is currently open, so the usual answer is one confirmation. "I don't know / not recorded" is a valid answer and is recorded as such. That is deliberate: a required field that can only be satisfied by naming a lot gets satisfied by naming the wrong one, and a confident wrong lot is worse than an honest gap (§8, and the same principle as the retro sentinel in §4). The same buy-off asks Materials used for each BOM line (CS-002 §7.4 step 4). A lot picked by scanning its label is recorded as scanned, so a checked lot and a remembered one read differently later.
5. Link the part to its mold records (`MOLD-SNx-###`) in the part's Mold section; a split mold is two links. Each run points at the one mold it was laid in, filled from the part's first mold; a run on the other half of a split mold is its own run. The mold's location, sealing record and Parts pulled count live on the mold, where they stay true across every work order that uses it, instead of being copied into each one.
6. Update the mold's location whenever it physically moves (PP-10). Scanning the mold and then the shelf is the fastest way and needs no typing.

7.3 Close

1. Fill the actuals in Quality (mass, thickness, Ω table, etc.), check the photos, and set the status to Complete. If an issue on the run is still open, the app opens the closeout list instead; dispose each one (§7.6).
2. Nothing to export by hand: every save writes straight to the shared database and is live for the whole team immediately. The durable archive is the lead's monthly Backup (header button; on a phone, Backup database in the menu) into the team Drive: Firestore is reliable, but a plain JSON file nobody can lock the team out of is the backup that matters. Restore, lead-only, reads that file back.
3. Print one copy for the physical binder at RFS if the part is safety/inspection-relevant (mech tech evidence).

7.4 Weekly use

1. At the weekly meeting the lead brings the status board (Reports, Print status board) or shows it on screen, and clears the Dashboard's Stopped lane first: it lists every run held at a blocker nobody has signed. The board replaces re-asking "where is X?" and feeds the weekly summary.
2. The lead runs Backup (one click, header; on a phone, in the menu) monthly; see §7.3. Nothing weekly is required since the database itself is always current; monthly is just the archive cadence.

7.5 Layup techniques (the step checklists)

A technique is a checklist and the blockers and holds on it: each step's rule (blocker, starts the cure, or the cure hold), the evidence it needs and the training it needs to sign. A part's Layup type picks its technique, and a run copies the steps when it is created.

1. Anyone can read the techniques on the Reports tab under Layup techniques.
2. Only a lead changes them. Add a technique always starts from a copy (Name, Start from); Edit steps changes one; Archive hides a custom technique from new runs. Built-in techniques are never archived.
3. Editing a technique never changes a run that already exists. A run made on an older version says its checklist has changed and offers Add them for the new steps, touching nothing already signed.
4. A technique that lays up a stack shall keep its Stack frozen blocker ahead of the first physical step, and a technique with a drop-test blocker shall keep it ahead of the resin step. The step editor lets a lead remove either; this rule is what stops it. Mold wet lay and Foam wrapped have no drop-test step; CS-007 §7.5 says why a lead does not add one there and where the readings go instead.
5. A lead who adds, edits or archives a technique shall open a revision of this standard within two weeks, so the table below catches up (CS-000 §2). The app holds the live checklists; this table is their value at this revision.

The built-in techniques as the app ships them (`app/workorders.js`, STD_STEPS, 2026-10-10). Blockers, holds, evidence and training are in brackets.

Technique	Layup type	Steps in order
Mold infusion	MOLD INFUSION	Stack frozen (blocker; needs a layup stack). Mold design review (blocker; needs the CAD; Mold design). Glue mold stock (ShopSabre CNC). Machine mold (needs a note; ShopSabre CNC). Seal and release mold. Dry stack and bag (Resin infusion). Drop test, 1 inHg or less over 10 min (blocker; needs a note; Resin infusion; the number is CS-006 §7.4's). Infuse (starts the cure; Resin infusion). Cure and demould (cure hold). Trim and finish
Glass infusion	GLASS INFUSION	Stack frozen (blocker; needs a layup stack). Prepare plate and release. Dry stack and bag (Resin infusion). Drop test, 1 inHg or less over 10 min (blocker; needs a note; Resin infusion). Infuse (starts the cure; Resin infusion). Cure and demould (cure hold). Cut to DXF, confirm revision. Finish
Mold wet lay	MOLD WET LAY	Stack frozen (blocker; needs a layup stack). Mold design review (blocker; needs the CAD; Mold design). Glue and machine mold (needs a note; ShopSabre CNC). Seal and release mold. Wet layup and bag (starts the cure; Wet layup). Cure and demould (cure hold). Trim and finish
Foam wrapped	FOAM WRAPPED	Stack frozen (blocker; needs a layup stack). Shape foam core (Foam core). Wet layup over core (starts the cure; Wet layup). Cure (cure hold). Trim and finish
Other	(blank, or any type no technique claims)	Define acceptance criterion: target and method, set before work starts (blocker; needs a note). Execute. Verify against criterion

7.6 Issues: raising and closing them

1. An issue shall be raised on the run it holds up: Raise an issue at the foot of the run's Issues section, or (Report an issue on this step) on the step where it happened.
2. Close it in the run's Issues section: Disposition (UAI (Use As Is), Corrective Action, Rework, Scrap or Other), What happened (the root cause), What was done, then Resolve. The issue's own page has the same three, under What happened, the disposition prompt, and Resolve, with the Resolve issue

button.

3. A run shall not complete while any of its issues lacks a Disposition, What happened or What was done. The app enforces Disposition and What was done: setting the run Complete opens the closeout list, with Resolve on each issue and Resolve all & complete work order at the foot. It does not check What happened, so the person closing out does.
4. A false alarm is cancelled with Cancel ticket (false alarm), not disposed.
5. If a fix did not hold, press Reopen on the issue's page. It withdraws the disposition and clears What was done, keeps What happened, and the run is held again until the issue is disposed a second time.

The What was done prompt follows the disposition: Scrap asks "Why could it not be saved?", UAI asks "Why is it acceptable as it is?".

7.7 Archiving, deleting and Recently deleted

1. A finished or abandoned part or run should be archived, not deleted. Any member can Archive one on its page in edit mode, or several with Select… on the list. An archived record leaves the lists, the Dashboard and the Season tab but keeps every field and who archived it; the archived chip on the list shows them, each with Restore.
2. Delete is for a record that should never have existed, and only a lead deletes parts, runs and issues. Deleting a run takes its issues and uploads with it.
3. A deleted record goes to Reports, Recently deleted, and can be put back with Restore for 30 days (`TRASH_DAYS` in `app/core.js`). Restoring brings back everything deleted with it.
4. After 30 days a lead presses Empty them, and the records are gone for good with their uploads.
5. A deleted ID is never reissued (§4.1 rule 2).

8. Quality checks / acceptance criteria

Check	Target	Method	Record where
Every part has a run before it consumes carbon or machine time	100%	The Work Orders tab lists Parts with no run yet; lead check at the weekly meeting	none
Blocker steps bought off in order	No downstream work past an unfinished blocker	App enforcement / shaded rows on paper	WO
Monthly Backup run	Every month during build season	Lead	Drive folder timestamps
Retro honesty	No fabricated buy-offs on retro WOs	"not recorded (retro)" convention	WO fields
R&D builds are marked as such	100%	Lead check at the weekly meeting: a part not in the season plan is either marked R&D or is a gap	Part `rnd` flag
Issues disposed and explained before Complete	100%	The app blocks Complete (closeout list, §7.6)	Run's Issues section
Steps that need training signed by trained people	Every override (training, evidence, cure hold) has a written reason	The app blocks it; the lead reads the override entries in the event logs at the weekly meeting	Run's Event log (in Notes & log)

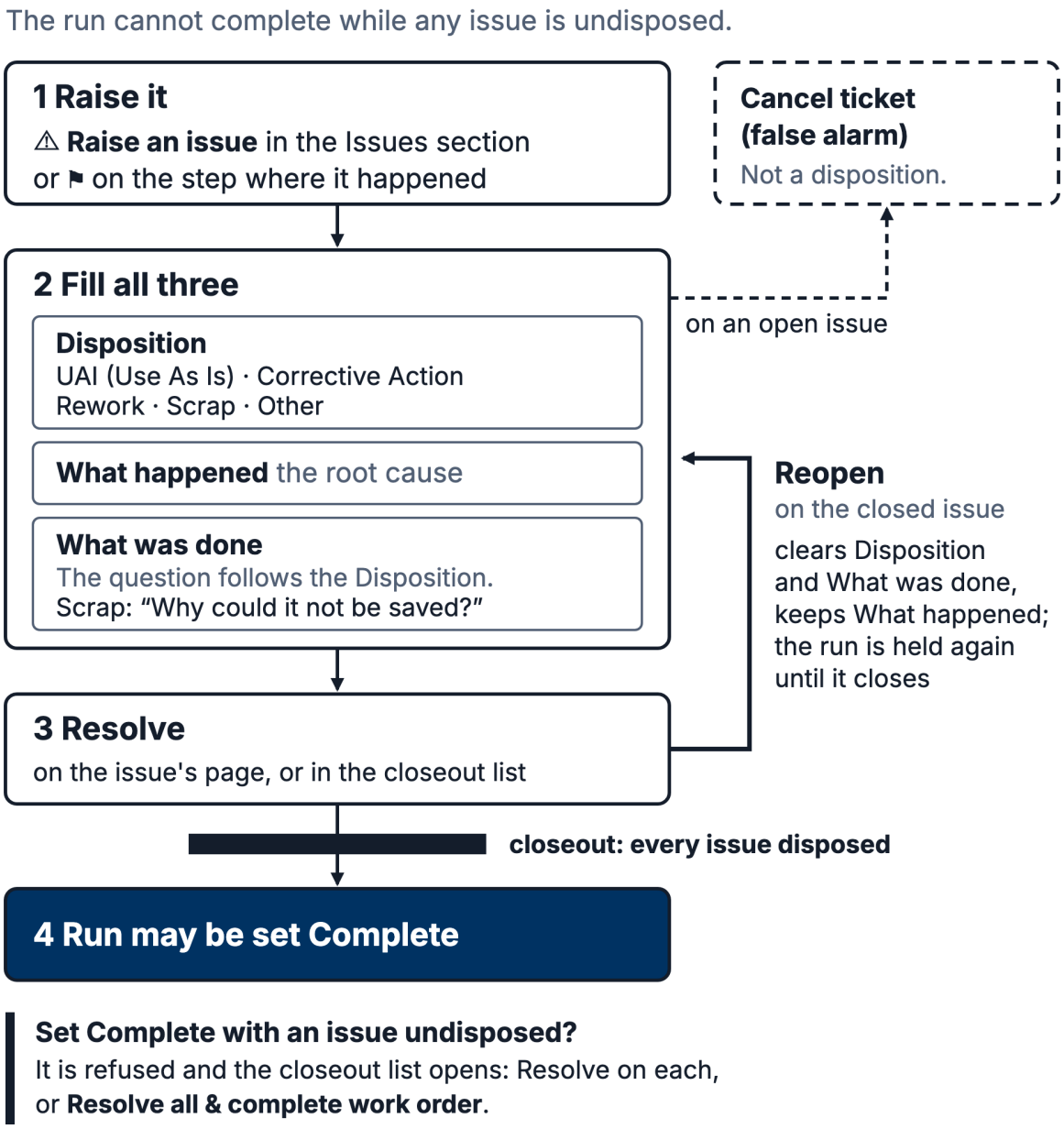


Figure 2: The issue path on a run: raise it (Raise an issue, or on a step), then Disposition, What happened and What was done; the app holds Complete until every issue has a Disposition and What was done. Reopen loops back and clears the disposition.

9. Records

The live database is the team's day-to-day part-history record, shared and current for everyone at once; the monthly Backup JSON in Drive is the archive nobody can lock the team out of. Master-Tracker-style views (spreadsheets, summaries) may continue as *views* built from that record. The WO is the source of truth (same rule as CS-002 §9).